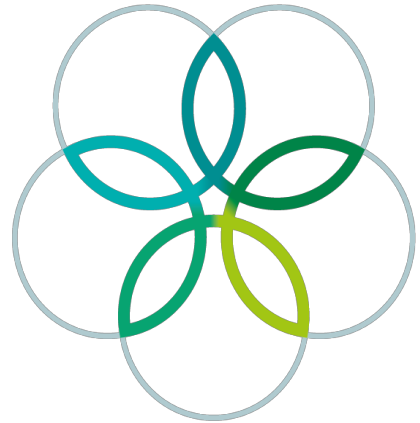
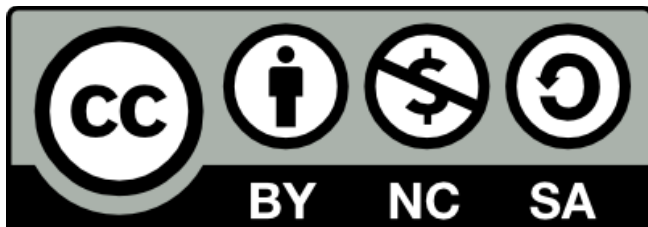


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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VI INTERNATIONAL BIOLOGY OLYMPIAD

1. Which characteristics are found in club-moss (*Lycopodium*) but not in horsetail (*Equisetum*)!

The codes of the answers:

1. spores have elaters;
 2. photosynthesis occurs in microphylls;
 3. sporophylls form strobilus;
 4. microphylls are arranged in a whorl. Answers:
- a) 1, 2; b) 2, 3; c) 2, 4; d) 3, 4.

2. Which is the first group of organisms that could successfully colonize a newly formed volcanic island?

- a) ferns; b) lichen; c) liverwort; d) algae.

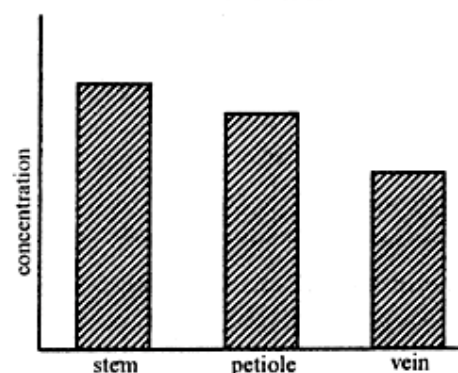
3. A population of mice originally inhabiting the entire area shown in the figure has become separated into two populations, A and B, by a new highway. If the environment inhabited by population A undergoes severe changes and the environment of population B does not. The rate of evolution of population A will probably be:

- a) initially slower than population B;
b) initially faster than population B;
c) equal to population B;
d) slower at first then faster than population B.



4. The concentration of ions in the sap of vessels of a tomato plant is investigated at three places (see the results in the figure). The difference in concentration of ions between vein and stem is partly attributed to:

- a) evaporation of water from the stomata;
b) capillary force of the xylem vessels;
c) intake of ions by leaf cells;
d) intake of water by leaf cells.



5. What is the most important function of glycolysis in aerobic cells?

- a) to obtain fat from glucose;
b) to obtain energy from glucose step by step;
c) to allow carbohydrates to enter the Krebs cycle;
d) ability to divide the glucose molecule into two pieces.

6. The frequencies of recombination between genes (loci) a, b, c, d, e and f linked on the same chromosome are: (a-c) 2,5%, (f-d) 8,5%, (b-d) 4,5%, (d-e) 4%, (c-e) 9,5%, (a-b) 20,5%, (f-a) 7,5%. The order of these genes (loci) is:

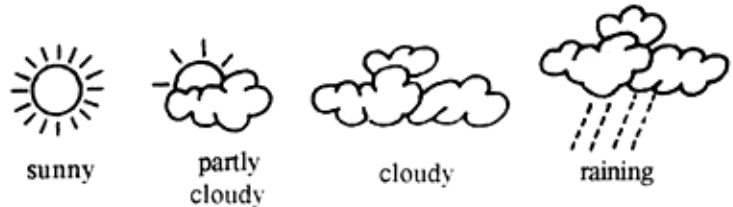
- a) a, c, d, e, f, b; b) b, c, e, f, d, a; c) a, c, f, e, d, b; d) b, e, f, c, a, d.

7. Cells of the adrenal cortex produce hormones whose structure is similar to that of:

- a) hemoglobin; b) cholesterol; c) tyrosine; d) adrenalin.

8. In green plants, which event can continue in all four conditions shown below?

- a) increasing net photosynthesis;
b) water absorption;
c) respiration;
d) transpiration;
e) guttation.



9. What type of behavior is shown when parents herring gull give an alarm call and the young birds respond by hiding?

- a) imprinting;
b) conditioned reflex;
c) reaction to a sign stimulus
d) displacement activity.

10. Which characteristics belong to insects?

The codes of the answers:

- 1. a dorsal rope ladder nerve cord;**
- 2. malpighian tubules;**
- 3. open circulatory system;**
- 4. gas exchange via a trachea system.**

Answers:

- a) 1, 2; b) 2, 3; c) 1, 2, 4; d) 2, 3, 4.

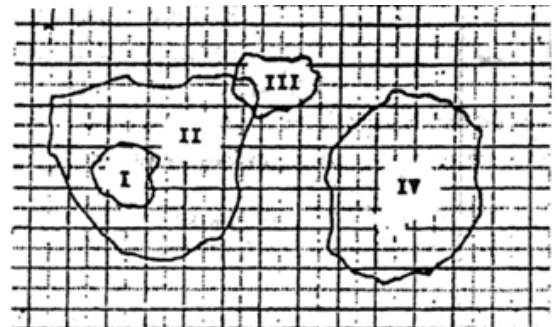
11. In which animal cells would the Golgi apparatus be most abundant?

- a) voluntary⁷ muscle cells; b) red blood cells; c) gland cells; d) ovums.

12. In the diagram, the background squares represent environmental factors (space, temperature, etc.), and the irregular polygons enclosing a set of factors represent the ecological niches of species I, II, III, and IV. Which species is in danger of elimination, if the resources are limiting?

Note: Each niche is continuous.

- a) I; b) II; c) III; d) IV



13. To determine that a green plant releases CO₂ during respiration, what is necessary in the experiment?

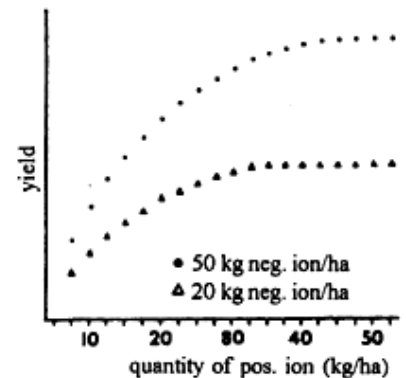
- a) using a plant with many leaves; b) doing the experiment in the dark;
c) submerging the plant in water; d) using a young plant.

14. Which of the following is an example of active transport?

- a) chloride exchange between red blood cells and plasma;
- b) sodium reabsorption in the distal tubules of the kidney;
- c) movement of oxygen from pulmonary alveoli into blood;
- d) oxygen movement within a muscle fiber.

15. The graph shows the relationship between the yield of a crop and the quantity of positive ions used to fertilize a field. A field is fertilized with 20 kg/ha of the positive ions and 20 kg/ha of the negative ions. Are the cations and anions under these conditions limiting factors for the yield?

- a) no, neither of them;
- b) only the positive ions;
- c) only the negative ions;
- d) yes, both of them.

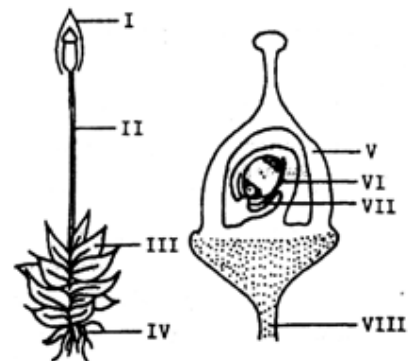


16. Incipient plasmolysis is the moment when:

- a) turgor pressure of the cell equals zero;
- b) the protoplast completely shrinks away from the cell wall;
- c) the cell volume is at a maximum;
- d) the cell wall can stretch no further.

17. Based on the figure above, identify plant parts of the same generation of the life cycle.

- a) III and VI, I and VI;
- b) I and V, II and VI;
- c) III and V, III and VI;
- d) II and VII, IV and VIII.



18. A condition necessary for speciation is:

- a) a high rate of gene mutation;
- b) geographical separation of populations;
- c) separation of a very small group of individuals from the initial large population;
- d) behavioural, geographical, genetic or other barriers preventing gene flow between populations.

19. The group of anamniotes is:

- a) pigeon, salamander, marsupial;
- b) dolphin, seahorse, seal;
- c) salmon, toad, skate;
- d) raven, woodpecker, newt.

20. In a gene pool with equal proportions of a dominant and a recessive alleles, complete selection against the recessive phenotype in each generation would:

- a) make little difference to the proportions of the genotypes;
- b) decrease the proportion of the recessive genotype;
- c) lead to the extinction of the recessive alleles;
- d) increase the proportion of heterozygotes.

21. Which of the following characters could be found in sea anemones and some sponges?

The codes of the answers:

- 1. pseudocoelom;**
- 2. intracellular digestion;**
- 3. radial symmetry;**
- 4. gastrovascular cavity.**

Answers: a) 1, 2; b) 2, 3; c) 3, 4; d) 1, 4.

22. Which of the following statements is not correct?

- a) phosphorylation of ADP occurs in the thylakoid membrane;
- b) ATP is formed as protons diffuse through ATP synthesis;
- c) ATP is consumed during the dark reaction;
- d) NADPH and ATP are produced in photosystem II.

23. Which of the following is observed when the growth rate of the population equals zero?

- a) the population is increasing and a strong competition for food and shelter is expected;
- b) the population is increasing and high parasitic and predatory activities are expected;
- c) the population is decreasing because of the accumulation of toxic waste;
- d) the population is near its carrying capacity.

24. A mouse was allowed to breathe air containing a particular isotope of oxygen $^{18}\text{O}_2$. In the mouse the "labeled" oxygen atoms first showed up in:

- a) pyruvate; b) carbon dioxide; c) acetyl-CoA; d) water.

25. Which abiotic factors limit the distribution of life in the ocean, but do not usually limit the distribution of life on land?

The codes of the answers:

- 1. minerals; 3. nitrogen;**
- 2. light; 4. oxygen.**

Answers:

a) 1, 3; b) 1, 4; c) 2, 3; d) 2, 4.

26. In higher plants, the evolution of the sporophyte has clearly dominated over that of the gametophyte (according to size, anatomic complexity and duration within the plant's life cycle). The primary reason for this dominance is that the sporophyte:

- a) may reproduce vegetatively; b) has a well-developed parenchyma;
- c) has a well-developed conducting tissue; d) has cells that divide by mitosis.

27. What would be observed when a grazer is removed from the ecosystem of a natural grassland?

The codes of the answers:

- 1. an increase in the intensity of plant competition;**
- 2. a decrease in the intensity of plant competition;**
- 3. an increase in the variety of plant species;**
- 4. a decrease in the variety of plant species.**

Answers:

a) 1, 3; b) 1, 4; c) 2, 3; d) 2, 4.

28. In which of the following events is crossing-over likely to occur?

- a) formation of spermatogonia;
- b) formation of spores in a fern;
- c) formation of egg in a liverwort archegonium;
- d) formation of a second plant from a strawberry stolon.

29. What is the immediate source of the energy used to make most of the ATP in animal cells?

- a) the transfer of phosphate groups from glucose breakdown products to ADP;
- b) the movement of hydrogen ions through a specific membrane;
- c) the splitting of glucose into two molecules of pyruvic acid;
- d) the movement of electrons along the electron transport chain.

30. In a seed, the food storage tissues for the embryo are:

- a) haploid in gymnosperms, triploid in angiosperms;
- b) diploid in gymnosperms, triploid in angiosperms;
- c) diploid in gymnosperms, pentaploid in angiosperms;
- d) haploid in gymnosperms, diploid in angiosperms.

31. Two cylinders P and Q are cut from a potato. P is placed for 1 hour in distilled water, and Q is placed for 1 hour in a salt solution with an osmotic value which is identical to the average value of cell sap of the potato cells. Determine whether the treated cylinders match their original holes in the potato.

- a) P does not match, but Q does;
- b) P does not match and neither does Q;
- c) P matches exactly and so does Q;
- d) P matches exactly, but Q does not.

32. Select characteristics specific for class Mammalia only.

The codes of the answers:

- 1. 4-chambered heart;
 - 2. sweat glands;
 - 3. diaphragm;
 - 4. homeothermy;
 - 5. pinna;
 - 6. scrotum;
 - 7. hair;
 - 8. viviparity.
- Answers:**
a) 1, 2, 4, 5; b) 3, 6, 7, 8; c) 2, 3, 5, 6, 7; d) 1, 4, 6, 7, 8.

33. A cross between two types of white-flowered sweet peas produced all purple-flowered peas in F_1 . 382 purple-flowered and 269 white-flowered peas were observed in F_2 . These numbers are consistent with the 9/7 ratio. If the purple F_1 were crossed to one of the parental types, what proportion of white-flowered peas would you expect among the progeny?

- a) 1; b) 0,75; c) 0,5; d) 0,25; e) 0.

34. One of the negative consequences of the overuse of antibiotics is:

- a) adaptation of the person undergoing treatment to increasing concentrations of the drug;
- b) stimulation of the production of antibodies;
- c) selection of antibiotic-resistant bacterial strains;
- d) increased frequency of mutations, eventually causing cancer.

35. Which characteristics do sunflowers have?

	Inflorescence	Corolla	Number of stamens per floret	Number of seeds per floret
a)	umbel	free	2	2
b)	head	united	5	1
c)	head	free	5	many
d)	umbel	united	5	1

36. U is inserted between the 9-th and 10-th base (counting in 5' - 3' direction) of the following mRNA: 5' GCUAUGCGCUACGAUAGCUAGGAAGC 3' and when it is translated into a peptide, the length of the peptide chain is*:

- a) 4; b) 5; c) 8; d) 9. *Use genetic code table.

TABLE OF GENETIC CODE
(molecules of mRNA)

	U	C	A	G	
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA (terminator)	UGA (terminator)	A
	UUG Leu	UCG Ser	UAG (terminator)	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA GluN	CGA Arg	A
	CUG Leu	CCG Pro	CAG GluN	CGG Arg	G
A	AUU Ileu	ACU Thr	AAU AspN	AGU Ser	U
	AUC Ileu	ACC Thr	AAC AspN	AGC Ser	C
	AUA Ileu	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met (initiator)	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val (initiator)	GCG Ala	GAG Glu	GGG Gly	G

37. The F₁ genotypes resulting from a cross between a drone honeybee and a¹ queen honeybee are males (AB, Ab, aB, ab) and females (AaBb, Aabb, aaBb, aabb). What are the genotypes of the parents?

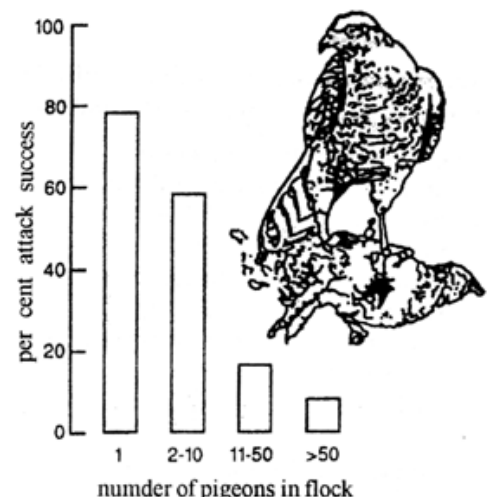
- a) aaBb x Ab; b) AaBb x ab; c) Aabb x aB; d) AaBb x Ab.

38. The following graph represents the probability of capture of a wood pigeon (*Columba palumbus*) by a goshawk (*Accipiter gentilis*) as a function of the size of the flock. Which of the above propositions are correct?

The codes of the answers:

1. a solitary wood pigeon has less chance of being captured by a goshawk than a pigeon in a flock;
2. the goshawks are less successful when they attack larger flocks of wood pigeons;
3. the goshawks attack only solitary wood pigeons;
4. the per cent attack success is inversely proportional to the number of pigeons in the flock. Answers:

- a) 1, 3; b) 1, 4; c) 2, 3; d) 2, 4.



39. Which is the typical characteristic of an Old World monkey?

- a) having a flat nose;
- b) lacking a prehensile tail;
- c) always having a long tail;
- d) exclusively ground dwelling.

40. The hypothesis postulated by A.Oparin and experimentally tested by S.Miller suggests that:

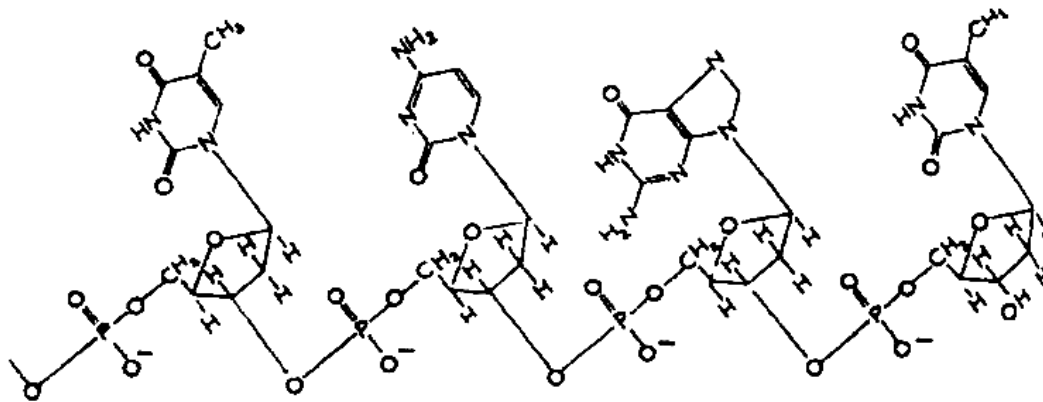
- a) the primitive atmosphere contained molecular oxygen;
- b) the primitive oceans contained high concentrations of proteins and nucleic acids;
- c) bacteria appeared on the earth $3,5 \times 10^9$ years ago;
- d) organic molecules could have been formed without life.

41. The chief role of ATP in neurotransmission is to:

- a) inhibit transport of Na^+ and K^+ across the membrane;
- b) induce an action potential;
- c) increase an action potential when it is already formed;
- d) maintain the resting potential.

42. The above nucleotide chain is:

- a) DNA; b) mRNA; c) tRNA; d) rRNA.



43. If frog tadpoles receive insufficient iodide from food and the surrounding water medium, which of the following may occur?

The codes of the answers:

- 1. enlargement of thyroid gland;
 - 2. over-secretion of TSH;
 - 3. growth stimulation;
 - 4. exhibition of cretinism;
 - 5. remaining in larval stage;
 - 6. enlargement of the pituitary gland. Answers:
- a) 1, 2, 3; b) 3, 4, 6; c) 2, 4, 6; d) 1, 2, 5.

44. Which of the following is typical for both gymnosperms (*Pinophyta*) and angiosperms (*Magnoliophyta*)?

- a) sporophylls differentiating into a carpel and a stigma;
- b) haploid endosperm and vascular tissues with tracheids;
- c) heterospory and nonflagellated sperm (male gamete);
- d) isogamy and wind pollination.

45. Which combination of the following human gametes will produce a Down syndrome male individual?

The codes of the answers:

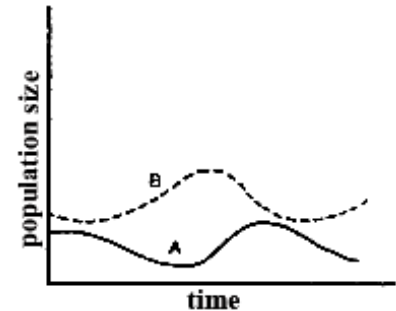
1. 23+X; 2. 21+Y; 3. 22+XX; 4. 22+Y.

Answers:

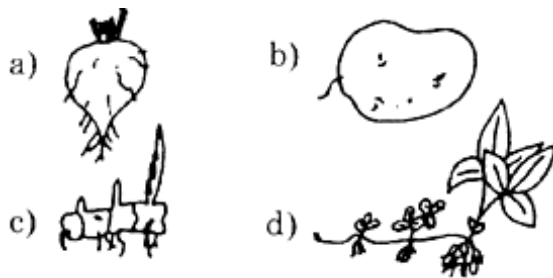
a) 1 and 2; b) 1 and 3; c) 1 and 4; d) 2 and 3; e) 3 and 4.

46. If A in the graph represents a population of hawks in a community, then what would most likely be represented by B?

- a) a population of the hawks' predators;
- b) a population with which the hawks have a mutualistic relationship;
- c) variation in the numbers of producers in that
- d) a population which is the hawks' prey.



47. Which of the following cannot reproduce asexually?



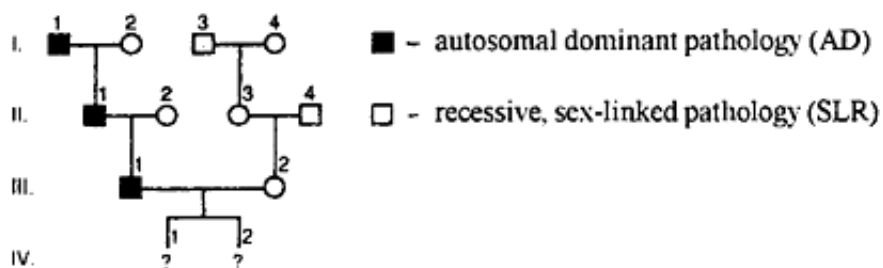
48. One locus has 5 alleles: $A_1, A_2, \dots, A_5$. How many different genotypes can exist in a population if the dominance hierarchy of these alleles is $A_1 > A_2 > A_3 > A_4 > A_5$?

a) 5; b) 10; c) 15; d) 32.

49. An average of 50 yeast cells per unit area was observed under the microscope. After 4 hours the liquid culture was diluted 10 times. Again a microscopic slide was prepared under the same conditions as before. An average of 80 cells per unit area was observed this time. What was the average time between cell divisions?

a) 1/4 hour; b) 1/2 hour; c) 1 hour; d) 2 hours.

50. Consider the pedigree below. If IV-1 is male and IV-2 is female, which of the following statements is correct?



- a) the probability that IV-1 would have both AD and SLR abnormalities is 1/8;
- b) the probability that IV-2 would have both AD and SLR abnormalities is 1/4;
- c) the probability that IV-1 would manifest AD abnormality but not the SLR abnormality is 1/8;
- d) the probability that IV-2 would manifest AD abnormality but not the SLR abnormality is 1/8.

51. Which are the possible conditions that could lead to serious hypoglycemia (low blood glucose level) and unconsciousness?

The codes of the answers:

1. type I diabetic patients (insufficient B-cells) who receive an insulin injection several hours before a meal;
2. type II diabetic patients (non-functional insulin receptors) who receive an excessive insulin injection;
3. patients with a tumor of the islets of Langerhans who receive an acute injection of insulin;
4. injection of insulin to a normal subject after heavy exercise.

Answers:

a) 1, 3; b) 1, 4; c) 1, 2, 3; d) 2, 3, 4.

52. Through how many of membranes would a molecule have to pass from the interior of a chloroplast thylakoid to the mitochondrial matrix?

a) 3; b) 5; c) 7; d) 9.

53. Substances can be transported across a membrane against their concentration gradient because:

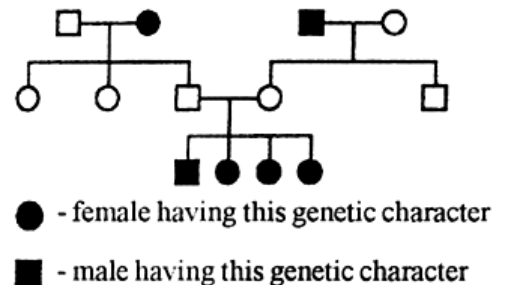
- a) some membrane proteins are ATP-dependent carrier molecules;
- b) some membrane proteins act as channels through which specific molecules can enter the cell;
- c) the lipid bilayer is permeable to numerous small molecules;
- d) the lipid bilayer is hydrophobic.

54. Of the following modes of inheritance, which one could describe the genetic character appearing in the above pedigree?

The codes of the answers:

1. autosomal dominant;
2. autosomal recessive;
3. sex-linked dominant;
4. sex-linked recessive.

Answers:
a) 1; b) 2; c) 1 or 3; d) 2 or 3; e) 2 or 4.



55. Which of the following is true for RNA?

- a) $G + C = A + U$;
- b) $G + C = C + U$;
- c) $G + C > A + U$;
- d) none of the above.

56. Which of the following numbers (lines) correctly matches stimuli specific for receptor cells A, B and C?

	A - hair cells in cochlea	B - olfactory epithelium	C - rod cells
a)	change in gravity	change in air	pressure light
b)	touch	change in chemical	heat
c)	vibration	change in chemical	light
d)	stretch	change in air	pressure electric current

57. If the following DNA is transcribed in the direction shown. The RNA product will be:

5' GCATTCGCCGA 3'
3' CGTAAGCGGCT 5'
←

- a) 5' UCGGCGAAUGC 3'; b) 5' GCAUUCGCCGA 3';
- c) 5' CGUAAGCGGCU 3'; d) 5' AGCCGCUUACG 3'.

58. A suitable vector for inserting DNA into the genome of a human cell is:

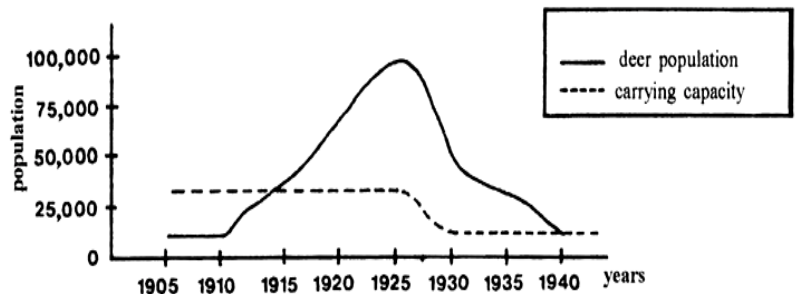
- a) T-plasmid; b) phage; c) retrovirus; d) all of the above.

59. Touching the mantle of the siphon of the seahare (*Aplysia*, phylum *Mollusca*), normally triggers a reflex that protects the mantle by withdrawing it. If the mantle is touched repeatedly, the withdrawal response becomes progressively weaker. This type of behaviour is called:

- a) habitation; b) a conditioned reflex;
c) trial and error; d) a chain of reflex.

60. The graph below depicts changes in the population growth rate of the Kaibab deer. About how many deer could this particular environment have supported in 1930 without some of them starving to death?

- a) 12000; b) 35000; c) 50000; d) 100000



61. A given fungus fails to digest starch in a certain culture medium. What are possible causes of this lack of digestion?

The codes of the answers:

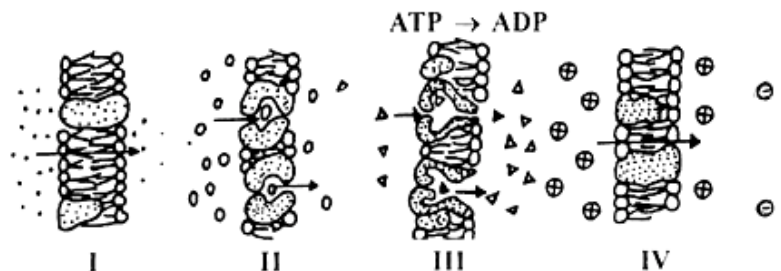
1. this fungus contains no amylase;
2. the amylase in the fungal mycelium is not secreted;
3. there is some substance interfering with starch digestion by the fungus;
4. the only respiratory substrate for this fungus is carbohydrate.

Answers:

- a) 1, 2; b) 3, 4; c) 1, 2, 3; d) 1, 2, 3, 4.

62. The figures I-IV illustrate transportation of substances and ions through the cell membranes. Which of the following statements is correct?

- a) there is diffusion in all figures;
b) there is active transport in all figures;
c) there is active transport in fig. II and III and passive transport in fig. I and IV;
d) there is osmosis in fig. I, II and IV;
e) there is active transport in fig. III and passive transport in fig. I, II, and IV.



63. Which of the following belong to dicots (*Magnoliopsida*)?

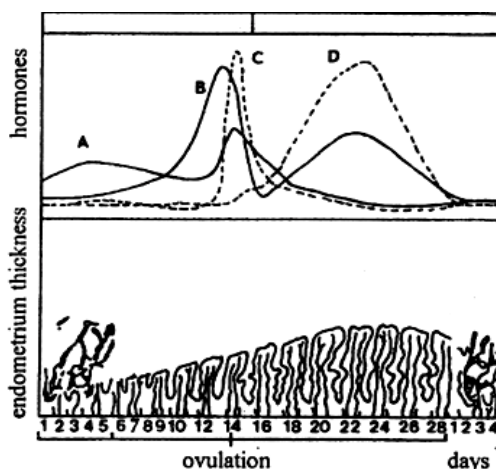
- a) banana, coconut, cucumber;
b) watermelon, cabbage, eggplant;
c) pineapple, onion, asparagus;
d) poppy, hemp, agave.

64. Which of the following is not important for migrating birds in finding and determining routes?

- a) auditory stimulation;
- b) infrared sensitivity;
- c) rotational force of the Earth;
- d) using the stars as a compass.

	B	C	D
a)	estradiol	progesterone	LH*
b)	LH	progesterone	estradiol
c)	estradiol	LH	progesterone
d)	LH	estradiol	progesterone

*LH – hormon of luteinization.



Four major reproductive hormones measured from blood serum of a woman during a normal menstrual cycle are shown in the figure. If A is FSH, what are B, C and D? Which is the most suitable condition to store seeds of most tropical plants so that they can remain viable for the longest time?

- a) in an ordinary refrigerator at 5°C;
- b) in a chamber at 5°C with 10% oxygen;
- c) in a chamber at 5°C with reduced pressure;
- d) in a chamber at 30°C with humidity maintained at 20%.

67. The above data show bacterial growth in various media (S.C.M. - simple culture medium, and U, V, X, Y, Z - represent different materials added to the medium). Which material cannot the bacteria synthesize?

- a)U; b)V; c)X; d)Y; e)Z.

<i>culture medium</i>	<i>growing state</i>
S.C.M.	-
S.C.M. + V + Z	-
S.C.M. + U + Y	+
S.C.M. + Z + X	-
S.C.M. + V + Y	+
S.C.M. + U + X	-
S.C.M. + Y + Z	+

68. Which of the following are not the characters of xerophytic plants?

The codes of the answers:

- 1. short stem;
- 2. stomata present on both sides of leaf surfaces;
- 3. enlargement of leaf surface;
- 4. hypoderma present.

Answers:

- a) 1,2; b)2,3; c) 3, 4; d) 1, 4.

69. Which codon can be mutated by one base change to a "nonsense" codon? (Use genetic code table on the p. 147).

- a) GCC; b) GAA; c) GCA; d) GGC

70. Which enzyme is not normally found in human?

- a) DNA-polymerase; b) hexokinase;
c) chitinase; d) ATP-synthetase.

71. What is the anticodon corresponding to the codon 5' GUA 3'?

- a) 5' CAU 3'; b) 5' UTC 3»; c) 5' UAC 3'; d) 5' AUG 3».

72. The methods of agriculture used by human have created serious insect problems chiefly because these practices:

- a) increase the rate of deforestation;
b) provide concentrated areas of food for insects;
c) increase the effectiveness of insecticides over a long period of time;
d) encourage insect resistance to their natural enemies.

73. Which are the differences between a higher plant cell and an animal cell and their respective mechanism of cell division?

The codes of the answers:

1. division of centromere; 3. function of spindle;
2. division of cytoplasm; 4. presence of centrioles.

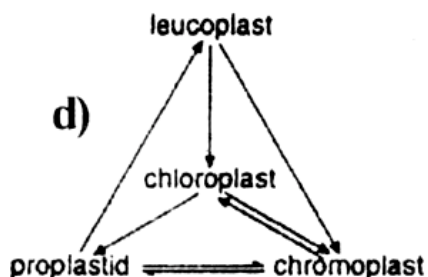
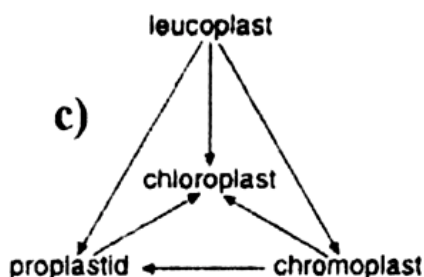
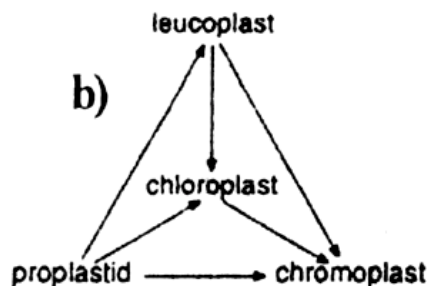
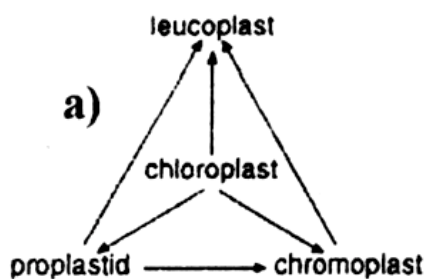
Answers:

- a) 1, 2; b) 1, 4; c) 2, 4; d) 3, 4.

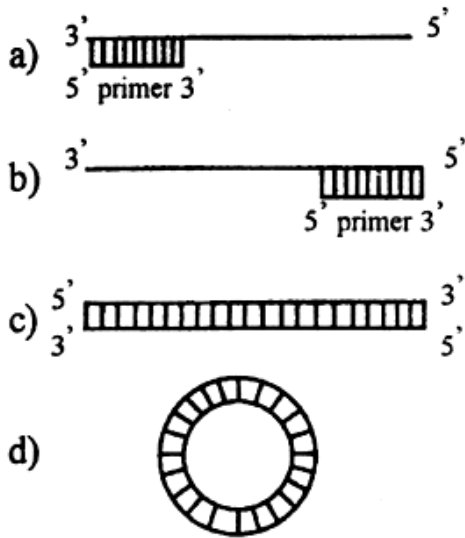
74. Which of the following is the common characteristic of reptiles, birds and mammals?

- a) teeth; b) diaphragm;
c) oxygenated blood is totally separated from deoxygenated blood;
d) kidneys are in the type of metanephros.

75. Plastides can develop from one another. Which figure is correct?



76. Which DNA is a substrate for DNA-polymerase?



77. Messenger RNA was transcribed in vitro from a double-stranded DNA molecule, which was later separated into single strands. For each strand of the DNA, the base ratio was analyzed and compared with that of mRNA. On the basis of the data given in the table, which strand of the double-stranded DNA served as the template for the mRNA synthesis?

	A	T or U	G	C
DNA-1	27,0	32,5	18,5	22,0
DNA-2	32,7	26,8	22,1	18,4
DNA-3	28,0	33,0	17,0	21,0
DNA-4	25,0	35,0	19,0	23,0
mRNA	27,0	33,0	17,0	23,0

a) DNA-1; b) DNA-2; c) DNA-3; d) DNA-4.

78. Which procedure would be used in a choice chamber to decide whether a response of animals to two different light intensities is taxis or kinesis?

- a) record the pathway of each animal;
- b) record the velocity of the animal movement;
- c) count, at intervals, the number of animals in each chamber;
- d) count, at intervals, the number of moving and stationary animals.

79. Which substances are found in pancreatic juice?

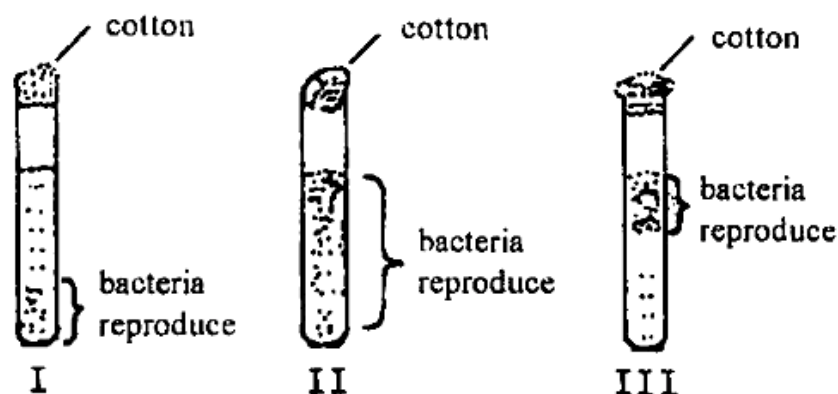
The codes of the answers:

- 1. bicarbonate;
- 2. secretin;
- 3. bile salts;
- 4. pepsinogen;
- 5. lipase.

Answers:

a) 1, 2; b) 1, 5; c) 2, 3, 5; d) 3, 4, 5.

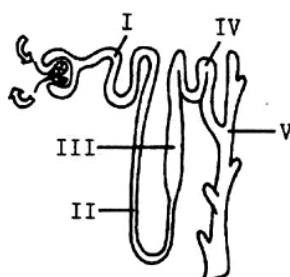
80. To determine their need for oxygen, three types of bacteria were grown in three different tubes. The regions of bacterial growth are shown in the figures above. Which is the correct classification of the bacterial types.



	Can reproduce only in aerobic medium	Can reproduce only in anaerobic medium	Can reproduce in every medium
a)	III	I	II
b)	III	II	I
c)	I	II	III
d)	I	III	II
e)	II	I	III

81. Which part of the nephron is impermeable to water?

a) I: b) II c) III: d) IV: e) V



82. A doctor has blood group 0, Rh⁻ and his wife A, Rh⁺. In an emergency situation the doctor quickly investigated the blood group of a patient. His own serum agglutinated with the patient's blood, but his wife's serum did not. What is the patient's blood group, and is it possible to conclude about his Rh-factor too?

	Blood group	Rh-factor
a)	A	yes
b)	A	no
c)	B	yes
d)	B	no

83. Which hormones increase and decrease blood glucose levels, respectively?

	Increase	Decrease
a)	triiodothyronine	insulin
b)	glucagon	Cortisol
c)	adrenalin	insulin
d)	oxytocin	adrenalin

84. Which of the following animal phyla possess giant axon systems?

- a) *Coelenterata*, *Plathelminthes*, *Nematoda*;
- b) *Plathelminthes*, *Nematoda*, *Annelida*;
- c) *Annelida*, *Arthropoda*, *Mollusca*;
- d) *Arthropoda*, *Mollusca*, *Chordata*.

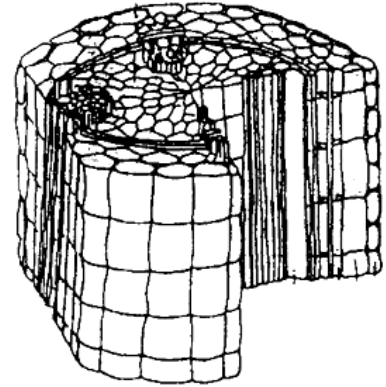
85. From the figure given below, choose the answer codes to fill in the blanks.

The codes of the answers:

- 1. *Polypodiophyta*;
- 2. *Magnoliophyta*;
- 3. *Pinophyta*;
- 4. primary growth;
- 5. secondary growth.

Answers:

- I. The figure is a plant stem in the division (s)
- II. The stage of growth is (are)



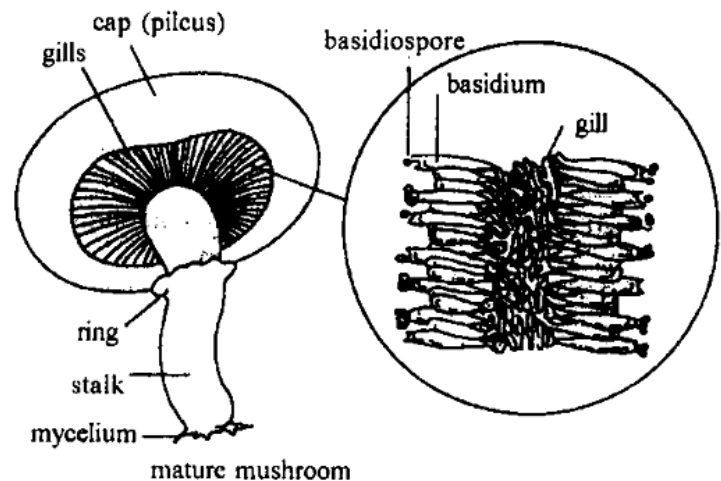
86. From the figure of a mushroom, choose the codes of the answers to answer the following questions.

The codes of the answers:

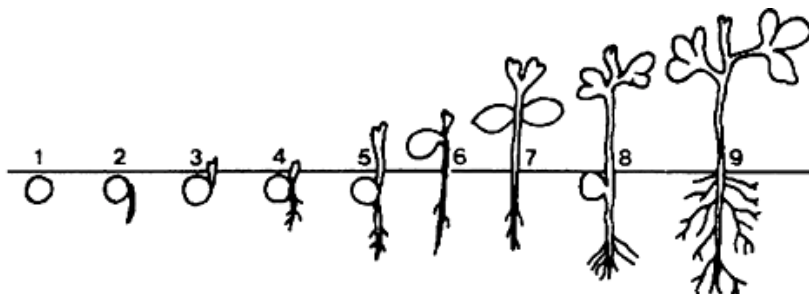
- 1. basidium;
- 2. basidiospore;
- 3. mycelium from basidiospore or primary mycelium;
- 4. mycelium from stalk or secondary mycelium;
- 5. zygote.

Answers:

- I. In which part of the mushroom does meiosis occur?
- II. Where is the diploid zygote formed?
- III. Which part of the mushroom is dicaryotic?



87. Select plant numbers from the diagram below and write a correct order of the phases of seed germination and development of young plants.



88. Put X in the appropriate place to indicate whether each statement concerning photosynthesis in plant is true or false.

- I. Photolysis occurs in photosystem I
- II. Oxygen is released
- III. NADH is formed

89. Fill in each blank with one correct answer code.

Name of plant tissue or organ	Origin	Function
collenchyma	ground meristem	support
chlorenchyma	(1).....	(2).....
primary xylem	(3).....	conducting
cork cambium	(4).....	producing cork
(5).....	(6).....	water absorption
pollen grain	(7).....	reproduction
pericycle	(8).....	origin of lateral roots

The codes of the answers:

- | | |
|---------------------|--------------------|
| 1. root cap; | 7. microspore; |
| 2. procambium; | 8. parenchyma; |
| 3. ground meristem; | 9. root hairs; |
| 4. primary xylem; | 10. protoderm; |
| 5. photosynthesis; | 11. transpiration; |
| 6. reproduction; | 12. aerenchyma; |
| | 13. respiration. |

90. Fill in the blanks with answer codes to indicate the main edible parts of the following fruits.

The codes of the answers:

- | | |
|----------------|---------------|
| 1. exocarp; | 5. perianth; |
| 2. mesocarp; | 6. endosperm; |
| 3. endocarp; | 7. cotyledon; |
| 4. receptacle; | 8. aril. |

Answers:

- I.Strawberry
 II.Banana
 III.Peanut
 IV.Orange
 V.Coconut

91. List all possible characteristics from the answer codes of the following organisms.

The codes of the answers:

- | | |
|-------------------------------|--|
| 1. autotrophic; | 6. having nucleus; |
| 2. heterotrophic; | 7. having main genetic materials in cytoplasm; |
| 3. chlorophyll in protoplasm; | 8. having cellulose cell wall; |
| 4. having chloroplasts; | 9. having mobility; |
| 5. having mitochondria; | 10. having no mobility. |

Answers:

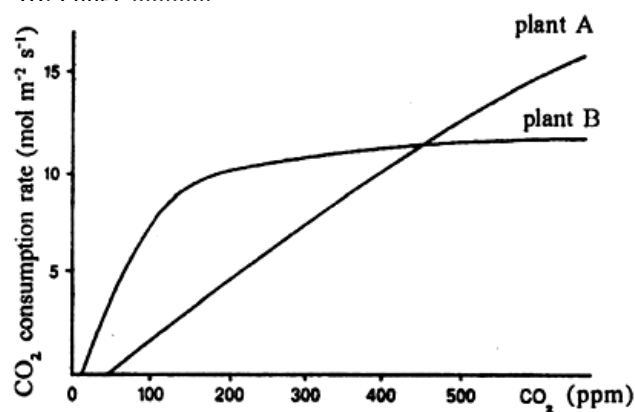
- I.Bacteria II.Blue-green algae III. Fungi

92. The figure shows the response of net carbon dioxide consumption rates of a single leaf to ambient CO₂ concentration (ppm) at a light intensity of 75% full sun.

Indicate with for true and "-" for false statements.

I. Plant A is a C₄-type because it has a net greater CO₂ consumption rate at high CO₂ concentration

II. At a CO₂ consumption rate equal to zero, there is no photosynthesis and respiration for both plants A and B



III. The CO_2 consumption rate of plant A will ultimately reach saturation as the CO_2 concentration increases further if light intensity is held constant...

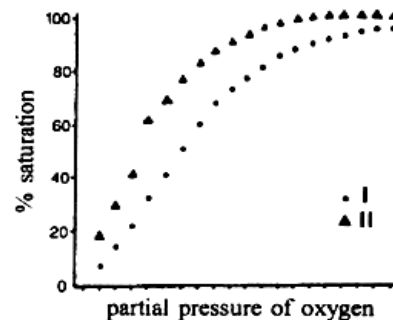
IV. C_4 -plants have greater light use efficiency in photosynthetic process than C_3 -plants at the CO_2 concentration of 200 ppm

V. At a light intensity of 100% full sun, the CO_2 consumption rate of plant B will reach saturation at CO_2 concentration greater than above in the graph

93. Saturation of blood hemoglobin with oxygen is influenced by the partial pressure of oxygen. This is investigated in four organisms: human adult, human fetus, llama (in the Andes) and cow. For two of them the results are shown in the figure. What could be the three possible pairs of I and II?

The codes of the answers:

code	I	II
1	human adult	llama
2	human fetus	llama
3	cow	llama
4	llama	human adult
5	human fetus	human adult
6	llama	cow
7	human adult	human fetus



Answers:

I. Possibility

II. Possibility

III. Possibility

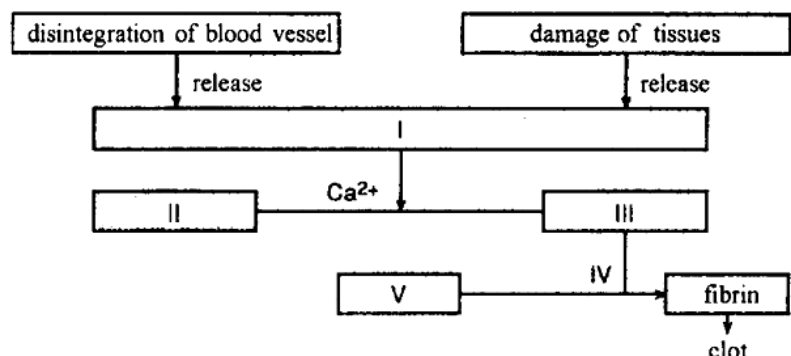
94. The following diagram shows the pathway of blood coagulation. What are I-V in the diagram?

The codes of the answers:

1. platelets;
2. thrombin;
3. thrombokinase;
4. fibrinogen;
5. prothrombin;
6. Ca^{2+} ;
7. Mg^{2+} ;
8. vitamin K.

9. Answers:

I.II.....III.....IV.....V



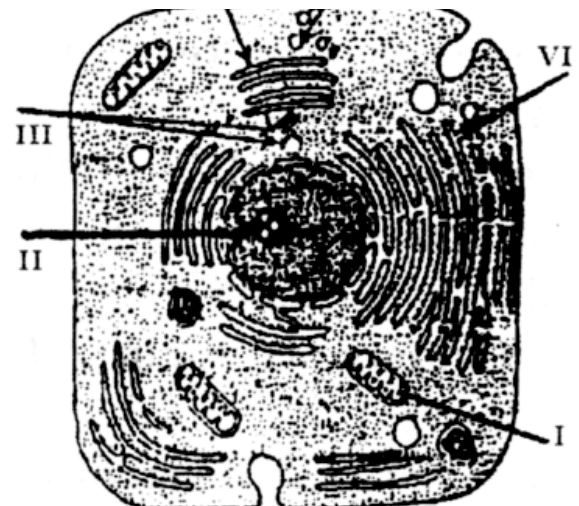
95. What are I-VI in the diagram?

The codes of the answers:

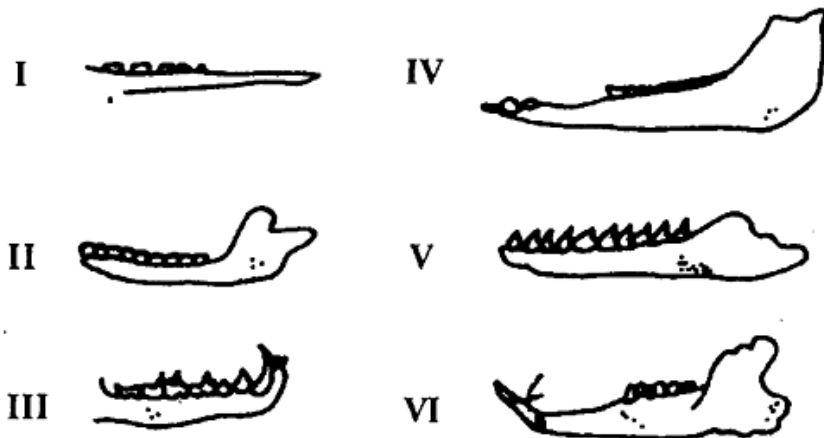
1. secretory vesicles;
2. microtubules;
3. ribosome;
4. lysosome;
5. mitochondrion;
6. vacuole;
7. chromosome;
8. Golgi body;
9. centriole;
10. nucleolus.

Answers:

I. ... II. ... III. ... IV. ... V. ... VI. .



96. Identify all lower jaws of mammals and non-mammalian vertebrates.



Answers:

I.Mammals

II.Non-mammalian vertebrates

97. Match all possible animal groups in column B with each description in column A:

I. no circulatory system;

II. open circulatory system;

III. closed circulatory system without heart chambers;

IV. heart with single-circuit circulation;

V. heart with 2 atria and 2 ventricles.

The codes of the answers (column B):

1. insect;

5. crocodile;

2. bird;

6. sea horse;

3. shark;

7. annelides (general);

4. planaria;

8. hydra.

Answers:

I II III IV V

98. Identify the following actions of the sympathetic and parasympathetic nervous systems.

The codes of the answers:

1. prepare body to cope with stressful condition;

2. dilate pupils;

3. increase motility of the intestine;

4. stimulate adrenalin secretion;

5. decrease heart rate;

6. stimulate oxytocin release.

Answers:

I. Sympathetic nervous system

II. Parasympathetic nervous system

99. The heron (*Ardeola ibis*) exhibits different types of nesting and breeding' which effect the success of offspring. The table shows the results:

Type of nesting (the codes of the answers)	Average number (per nest) of offspring flying out	Average amount of food brought to the nest per day
1	1,90	65,1 ml
2	2,29	53,1 ml
3	2,33	68,3 ml

With these data it is possible to compare the efficiency in energy investment of the parent birds. Which type of nesting (number 1, 2 or 3 from the table) has the lowest efficiency and which one has the highest?

Answers:

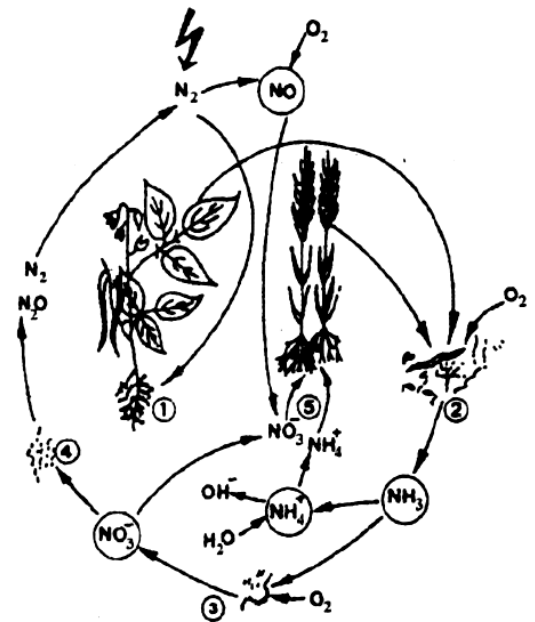
- I. The lowest efficiency
 II. The highest efficiency

100. While investigating and comparing the efficiency in energy investment of the parent birds, some conditions have to be considered. Choose two possible conditions below.

The codes of the answers:

1. all young birds have to be about the same size during the investigation;
2. all parent birds have to have about the same mass;
3. the nests have to be close together;
4. the food has to be at about the same distance from all the nests;
5. the experiment should last no longer than one year;
6. all the parent birds should feed their young on the same type of food.

Answers:



101. Choose the correct number (1-5) from the given diagram of the nitrogen cycle to match each statement (I-V).

Answers:

- I. Fixation of nitrogen by the bacteria in nodules
 II. Absorption of nitrogen compounds by the roots
 III. Action of nitrifying bacteria
 IV. Action of denitrifying bacteria
 V Action of decomposing bacteria

102. Assume the genotype AABb was crossed with recessive genotype and the F_1 offspring are mated at random to produce F_2 offspring consisting of 22 A-B-, 5 A- bb, 5 aaB- and 4 aabb.

Estimate the recombination frequency between A and B?

Answer:

103. Let "x" be the amount of nuclear DNA in a gamete of a diploid organism. Fill in each blank the amount of nuclear DNA in different stages of the cell cycle of this organism. The DNA amount per cell in the following stages of the cell cycle (example: 0,5x, x, 2x or 4x)

I. Mid S =

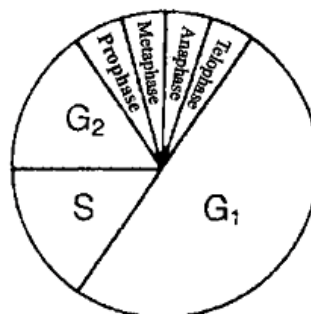
II. G_1 =

III. G_2 =

VI. Anaphase =

V. Metaphase =

IV. Prophase =



104. If the initial frequencies of genotypes A_1A_1 , A_1A_2 and A_2A_2 are 0,04, 0,32 and 0,64, respectively, what are the genotypic frequencies after one generation of complete self-fertilization?

Answers:

I. A_1A_1 II. A_1A_2 III. A_2A_2

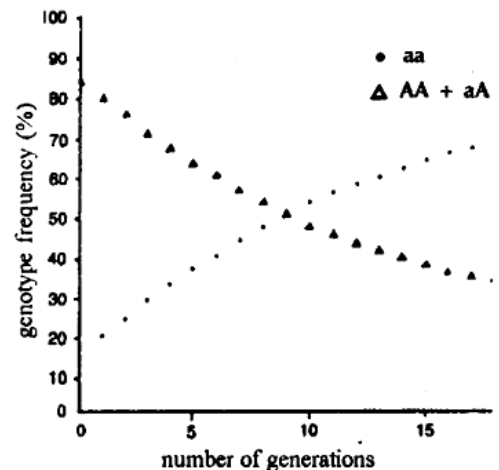
105. The diagram shows genotypic frequencies during subsequent generations in a population.

I. What was the frequency of allele A at generation 0, assuming a Hardy-Weinberg equilibrium at that point?

Answer: frequency of allele A is

II. What would be the frequency of aa after a very large number of generations?

Answer: frequency of aa would be



106. In a human population at equilibrium, the frequency of $I^A = 0,2$, $I^B = 0,4$ and $i = 0,4$. What are the genotypic frequencies of blood group A, B, AB and 0?

Answers:

I. Group A

II. Group B

III. Group AB

IV. Group 0

107. If the statement is correct for both chloroplasts and mitochondria, mark with "+". If it is not correct for both, mark with

I. Contain proteins

II. Contain coenzymes for binding hydrogen

III. Contain K^+ ions

IV. Lack DNA

V Can produce ATP

VI. Can produce oxygen

108. What would sequentially happen if the sea was polluted with a large number of organic substances? Write the correct sequence of such changes using the answer codes.

The codes of the answers:

1. the amount of oxygen in water decreases;

2. the bacteria decomposing the dead organisms breed rapidly;

3. the amount of oxygen in water increases;

4. there is the growth of bacteria producing H_2S ;

5. the planktonic algae reproduce rapidly.

Answer: the correct sequence is